

CSE370 : Database Systems

Assignment 02 | Spring 2025

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No 1 Query (as Plain Text)	SELECT first_name, last_name, email, phone_number, hire_date, department_id FROM Employee WHERE hire_date = (SELECT MAX(hire_date) FROM Employee);
No 1 SS (of Query & Output in Shell)	<pre>MariaDB [Company_22201344]> SELECT first_name, last_name, email, phone_number, hire_date, department_id -> FROM Employee -> WHERE hire_date = (SELECT MAX(hire_date) FROM Employee); +-----+-----+-----+-----+-----+-----+ first_name last_name email phone_number hire_date department_id +-----+-----+-----+-----+-----+-----+ Rashid Khan rashid.khan@email.com 12345678901243 2017-06-09 DPT003 +-----+-----+-----+-----+-----+-----+ 1 row in set (0.003 sec)</pre>
No 2 Query (as Plain Text)	SELECT first_name, last_name, employee_id, phone_number, salary, department_id FROM Employee e1 WHERE salary = (SELECT MIN(salary) FROM Employee e2 WHERE e1.department_id = e2.department_id);

No 2 SS
(of Query & Output
in Shell)

```
MariaDB [Company_22201344]> SELECT first_name, last_name, employee_id, phone_number, salary, department_id
-> FROM Employee e1
-> WHERE salary = (
->     SELECT MIN(salary)
->     FROM Employee e2
->     WHERE e1.department_id = e2.department_id
-> );
```

first_name	last_name	employee_id	phone_number	salary	department_id
Virat	Kohli	EMP001	12345678901234	50000	DPT001
Rohit	Sharma	EMP004	12345678901237	55000	DPT004
Babar	Azam	EMP005	12345678901238	45000	DPT005
Kane	Williamson	EMP006	12345678901239	48000	DPT006
Joe	Root	EMP007	12345678901240	52000	DPT007
David	Warner	EMP008	12345678901241	50000	DPT001
Ben	Stokes	EMP009	12345678901242	49000	DPT002
Rashid	Khan	EMP010	12345678901243	47000	DPT003
Anil	Kumble	EMP020	12345678901253	28000	DPT008
Ravindra	Jadeja	EMP021	12345678901254	40000	DPT009
Yuvraj	Singh	EMP023	12345678901256	37000	DPT010
Faf	du Plessis	EMP025	12345678901258	41000	DPT011

12 rows in set (0.001 sec)

No 3 Query
(as Plain Text)

```
SELECT first_name, last_name, employee_id, commission_pct, department_id
FROM Employee
WHERE department_id = 'DPT007'
AND commission_pct < ALL (SELECT MIN(commission_pct) FROM Employee WHERE department_id = 'DPT005');
```

No 3 SS (of Query & Output in Shell)	<pre> MariaDB [Company_22201344]> SELECT first_name, last_name, employee_id, commission_pct, department_id -> FROM Employee -> WHERE department_id = 'DPT007' -> AND commission_pct < ALL (SELECT MIN(commission_pct) FROM Employee WHERE department_id = 'DPT005'); +-----+-----+-----+-----+-----+ first_name last_name employee_id commission_pct department_id +-----+-----+-----+-----+-----+ Joe Root EMP007 0.030 DPT007 +-----+-----+-----+-----+-----+ 1 row in set (0.001 sec) </pre>
No 4 Query (as Plain Text)	<pre> SELECT department_id, COUNT(*) AS total_employees FROM Employee GROUP BY department_id HAVING MAX(salary) <= 30000; </pre>
No 4 SS (of Query & Output in Shell)	<pre> MariaDB [Company_22201344]> SELECT department_id, COUNT(*) AS total_employees -> FROM Employee -> GROUP BY department_id -> HAVING MAX(salary) <= 30000; +-----+-----+ department_id total_employees +-----+-----+ DPT008 2 +-----+-----+ 1 row in set (0.001 sec) </pre>
No 5 Query (as Plain Text)	<pre> SELECT e1.department_id, e1.job_id, e1.commission_pct FROM Employee e1 </pre>

	WHERE EXISTS (SELECT 1 FROM Employee e2 WHERE e1.department_id = e2.department_id AND e1.commission_pct < e2.commission_pct);
No 5 SS (of Query & Output in Shell)	<pre> MariaDB [Company_22201344]> SELECT e1.department_id, e1.job_id, e1.commission_pct -> FROM Employee e1 -> WHERE EXISTS (-> SELECT 1 -> FROM Employee e2 -> WHERE e1.department_id = e2.department_id -> AND e1.commission_pct < e2.commission_pct ->); +-----+-----+-----+ department_id job_id commission_pct +-----+-----+-----+ DPT001 JOB008 0.050 DPT002 JOB009 0.040 DPT003 JOB010 0.020 DPT009 JOB021 0.040 DPT010 JOB024 0.040 DPT011 JOB025 0.020 +-----+-----+-----+ 6 rows in set (0.001 sec) </pre>
No 6 Query (as Plain Text)	SELECT manager_id FROM Employee

	WHERE manager_id IS NOT NULL GROUP BY manager_id HAVING MIN(salary) >= 3500;
No 6 SS (of Query & Output in Shell)	<pre> MariaDB [Company_22201344]> SELECT manager_id -> FROM Employee -> WHERE manager_id IS NOT NULL -> GROUP BY manager_id -> HAVING MIN(salary) >= 3500; +-----+ manager_id +-----+ MNG001 MNG002 MNG003 MNG004 MNG005 MNG010 MNG011 MNG012 MNG013 +-----+ 9 rows in set (0.012 sec) </pre>
No 7 Query	SELECT first_name, last_name, employee_id, email, salary, department_id, commission_pct

(as Plain Text)

```
FROM Employee e1
WHERE commission_pct = (
  SELECT MIN(e2.commission_pct)
  FROM Employee e2
  WHERE e1.manager_id = e2.manager_id
);
```

No 7 SS
(of Query & Output
in Shell)

```
MariaDB [Company_22201344]> SELECT first_name, last_name, employee_id, email, salary, department_id, commission_pct
-> FROM Employee e1
-> WHERE commission_pct = (
->   SELECT MIN(e2.commission_pct)
->   FROM Employee e2
->   WHERE e1.manager_id = e2.manager_id
-> );
```

first_name	last_name	employee_id	email	salary	department_id	commission_pct
MS	Dhoni	EMP002	ms.dhoni@email.com	60000	DPT002	0.060
Rohit	Sharma	EMP004	rohit.sharma@email.com	55000	DPT004	0.050
Babar	Azam	EMP005	babar.azam@email.com	45000	DPT005	0.040
Joe	Root	EMP007	joe.root@email.com	52000	DPT007	0.030
Rashid	Khan	EMP010	rashid.khan@email.com	47000	DPT003	0.020
Rahul	Dravid	EMP019	rahul.d@email.com	29000	DPT008	0.010
Anil	Kumble	EMP020	anil.k@email.com	28000	DPT008	0.010
Ravindra	Jadeja	EMP021	ravindra.j@email.com	40000	DPT009	0.040
Shaun	Marsh	EMP024	shaun.m@email.com	38000	DPT010	0.040
Faf	du Plessis	EMP025	faf.d@email.com	41000	DPT011	0.020

```
10 rows in set (0.001 sec)
```